Extra Practice 1: Rational Expressions and Non-Permissible Values

A. Skills

1. Evaluate $\dfrac{x^2+1}{x-4}$ for each value of x.

(a) $x = 2$ (b) $x = -3$ (c) $x = 4$ (d) $x = \frac{1}{2}$

2. Determine the non-permissible values.

(a) $\dfrac{7}{2y}$ (b) $\dfrac{a-3}{a+11}$ (c) $\dfrac{1}{9-2x}$

(d) $\dfrac{x^2}{x^2+49}$ (e) $\dfrac{6}{c(c+8)}$ (f) $\dfrac{x+5}{6}$

3. Factor the denominator and determine the non-permissible values.

(a) $\dfrac{x}{x^2-8x+12}$ (b) $\dfrac{3}{4x^2-1}$ (c) $\dfrac{p-1}{5p^2+20p}$

(d) $\dfrac{1}{2x^2-x-15}$ (e) $\dfrac{x+4}{x^2+14x+49}$ (f) $\dfrac{2}{3x^2-27}$

B. Applying

4. Complete the table for $\dfrac{1}{x-5}$ and explain what happens near $x = 5$.

x	4.9	4.99	5	5.01	5.1
$\dfrac{1}{x-5}$					

5. Determine the non-permissible values for x in terms of y. (a) $\dfrac{2}{x+6y}$ (b) $\dfrac{x-y}{x^2-7xy+12y^2}$

6. Determine the non-permissible values of $\dfrac{3}{x-2} - \dfrac{x+1}{x^2+5x}$.

7. Write a rational expression with a numerator of 1 and the denominator expanded that has the given NPVs. (a) $x \neq 5, -2$ (b) $x \neq 0, \pm 1$ (c) $x \neq 0, \frac{3}{4}$

8. Students share the \$180 cost of a bus. If n students sign up but 3 drop out, each remaining student pays $\dfrac{180}{n-3}$ dollars.

(a) Determine the cost per student when $n = 12$, and how much more that is than if nobody had dropped out.

(b) State the NPV and the values of n that make sense.

9. The non-permissible values of $\dfrac{x^2-4}{x^2+4x}$ are

A. $x \neq \pm 2$ **B.** $x \neq 0, -4$ **C.** $x \neq 0, 4$ **D.** $x \neq -4$ only

10. Numerical Response The positive non-permissible value of $\dfrac{x-3}{6x^2-x-12}$, to the nearest tenth, is _____.

C. Stretch

11. The expression $\dfrac{x+1}{x^2-7x+k}$ has a non-permissible value of $x = 2$. Determine k and the other non-permissible value.

12. Determine the non-permissible values of $\dfrac{x}{x^4-16}$. Explain why there are only two.

13. Consider $\dfrac{1}{x^2-8x+k}$.

(a) For which value of k is there exactly one NPV? State it.

(b) For which values of k are there no NPVs? Explain.

Extra Practice 2: Simplifying Rational Expressions by Factoring

A. Skills

1. Simplify. State the non-permissible values.

(a) $\dfrac{15a^3b}{25ab^4}$ (b) $\dfrac{7(x-1)^2}{21(x-1)}$ (c) $\dfrac{12x-8}{4x}$ (d) $\dfrac{x^2-16}{5x-20}$

2. Simplify. State the non-permissible values.

(a) $\dfrac{x^2-x-20}{x^2+9x+20}$ (b) $\dfrac{y^2+6y+9}{y^2-9}$

(c) $\dfrac{3x^2+11x-4}{9x^2-1}$ (d) $\dfrac{4m^2+4m-24}{2m^2-8}$

3. Simplify. State the non-permissible values.

(a) $\dfrac{5-x}{x-5}$ (b) $\dfrac{16-x^2}{x^2+x-20}$ (c) $\dfrac{3a-6}{8-2a-a^2}$

B. Applying

4. Simplify, and state the non-permissible values for x in terms of y. (a) $\dfrac{x^2-16y^2}{x^2+4xy}$ (b) $\dfrac{6x-3y}{y^2-4x^2}$

5. A student writes $\dfrac{x^2+3x}{x^2+5x}=\dfrac{3x}{5x}=\dfrac{3}{5}$. Explain the error, show with a value of x that the answer is wrong, and simplify correctly.

6. Is $\dfrac{x^2-1}{x^2-2x+1}$ equivalent to $\dfrac{x+1}{x-1}$? Is $\dfrac{x^2+1}{x+1}$ equivalent to $x+1$? Explain.

7. A rectangle has an area of $(4x^2-9)$ m^2 and a width of $(2x-3)$ m.

(a) Write and simplify an expression for the length.

(b) The perimeter is 40 m. Determine x and the dimensions.

8. The simplest form of $\dfrac{2x^2-8}{x^2-x-2}$ is

A. $\dfrac{2(x+2)}{x+1}$ **B.** $\dfrac{2(x-2)}{x+1}$ **C.** $\dfrac{x+2}{x+1}$ **D.** $\dfrac{8}{x+2}$

9. Numerical Response Simplify $\dfrac{x^2+2x-15}{2x^2-5x-3}$ and evaluate it at $x=2$. To the nearest tenth, the value is _____.

C. Stretch

10. The expression $\dfrac{3x^2+kx-4}{x^2-4}$ simplifies to $\dfrac{3x+2}{x+2}$. Determine k.

11. Simplify $\dfrac{x^4-13x^2+36}{x^2+x-6}$. State the non-permissible values.

12. Simplify $\dfrac{a^2-b^2}{(b-a)^2}$, and state the non-permissible value for a in terms of b.

13. Write a rational expression with a trinomial numerator and a trinomial denominator that simplifies to $\dfrac{2x+1}{x-3}$ and has the NPVs $x \neq 3, -4$.

Extra Practice 3: Multiplying Rational Expressions

A. Skills

1. Simplify. State the non-permissible values where there are variables.

(a) $\frac{15}{14} \times \left(-\frac{21}{40}\right)$ (b) $\frac{10x^4y}{9z^2} \cdot \frac{3z^3}{25x^2y^2}$ (c) $\frac{-6p^2}{7q} \cdot \frac{14q^3}{9p^4}$

2. Simplify. State the non-permissible values.

(a) $\frac{x^2 - x}{x + 4} \cdot \frac{3x + 12}{x^2 - 1}$ (b) $\frac{7x + 14}{x^2 - 3x} \cdot \frac{2x - 6}{x^2 - 4}$

3. Simplify. State the non-permissible values.

(a) $\frac{x^2 + 6x + 8}{x^2 - x - 6} \cdot \frac{x^2 - 5x + 6}{x^2 + 3x - 4}$ (b) $\frac{3y^2 + 5y - 2}{y^2 - 4} \cdot \frac{y^2 - 2y}{9y^2 - 1}$

4. Simplify. State the non-permissible values.

(a) $\frac{4 - x}{x^2 + x} \cdot \frac{x^2 + 5x + 4}{x^2 - 16}$ (b) $\frac{x^2 - x - 12}{10 - 5x} \cdot \frac{x^2 - 4}{x^2 + x - 6}$

B. Applying

5. Simplify $\frac{x^2 - 25}{3x} \cdot \frac{6x^2}{x^2 + x - 20} \cdot \frac{x - 4}{x - 5}$. State the non-permissible values.

6. Simplify $\dfrac{m^2-4n^2}{m^2+mn} \cdot \dfrac{m^2-n^2}{m^2-mn-2n^2}$. State the restrictions on the variables.

7. A box (rectangular prism) has length $\dfrac{x+2}{x-1}$, width $\dfrac{x^2-1}{4x}$ and height $\dfrac{8x^2}{x^2+4x+4}$, all in metres, where $x > 1$.

(a) Write and simplify an expression for its volume.

(b) Determine the volume when $x = 3$.

8. For all permissible values of x, $\dfrac{2x-6}{x^2-x} \cdot \dfrac{x^2-1}{9-x^2}$ is equal to

A. $-\dfrac{2(x+1)}{x(x+3)}$ **B.** $\dfrac{2(x+1)}{x(x+3)}$ **C.** $-\dfrac{2(x+1)}{x(x-3)}$ **D.** $\dfrac{2}{x(x+3)}$

9. Numerical Response For all permissible values of x, the product $\dfrac{5x^2-20}{x^2-3x} \cdot \dfrac{x^2-9}{x^2+5x+6} \cdot \dfrac{2x}{x-2}$ simplifies to a whole number. The number is _____.

C. Stretch

10. Determine the polynomial that belongs in the box: $\dfrac{x^2+2x-15}{\square} \cdot \dfrac{x^2-4}{x+5} = \dfrac{x+2}{x}$

11. Simplify $\dfrac{x^2-2x}{x^2-1} \cdot \dfrac{x+1}{x^2-4}$, state the NPVs, then determine its exact value when $x = \sqrt{3}$, in simplest form.

12. Jordan says $\dfrac{x-1}{x+3} \cdot \dfrac{x+3}{x-1}$ is equal to 1 for every value of x. Is Jordan right? Explain, and describe how the graphs of $y = \dfrac{x-1}{x+3} \cdot \dfrac{x+3}{x-1}$ and $y = 1$ differ.

Extra Practice 4: Dividing Rational Expressions

A. Skills

1. Simplify. State the non-permissible values where there are variables.

(a) $-\frac{21}{25} \div \frac{14}{15}$

(b) $\frac{9x^4}{8y^3} \div \frac{3x}{4y}$

(c) $\frac{-5a^2b^3}{12c} \div \frac{10ab}{9c^4}$

2. State the non-permissible values of $\frac{x+4}{x^2-25} \div \frac{3x-2}{x^2+6x}$. Do not simplify.

3. Simplify. State the non-permissible values.

(a) $\frac{x^2-49}{5x} \div \frac{x+7}{10x^2}$

(b) $\frac{x^2+2x-15}{x^2-6x+8} \div \frac{x^2+8x+15}{x^2-16}$

(c) $\frac{4y^2-4y+1}{y^2+y} \div \frac{2y^2+5y-3}{y^2-1}$

(d) $\frac{x-6}{x^2+2x} \div \frac{36-x^2}{x^2-4}$

B. Applying

4. Simplify $\frac{x^2+x-2}{x^2-3x} \div \frac{x^2-1}{x^2-9} \cdot \frac{x}{x+3}$. State the non-permissible values.

5. Simplify $\frac{a^2-9b^2}{a^2+ab} \div \frac{a^2-2ab-3b^2}{a^2-b^2}$. State the restrictions on the variables.

6. Simplify $\dfrac{\dfrac{x^2-4x}{x+2}}{\dfrac{x^2-16}{x^2+2x}}$. State the non-permissible values.

7. A rectangular prism has volume $\dfrac{x^2+5x}{x-1}$ cubic metres and base area $\dfrac{x^2-25}{x^2-1}$ square metres, where $x > 5$.

(a) Write and simplify an expression for its height.

(b) Determine the height when $x = 7$.

8. The non-permissible values of $\dfrac{x+3}{x-2} \div \dfrac{x-4}{x+1}$ are

A. $x \neq 2$

B. $x \neq 2, -1$

C. $x \neq 2, -1, 4$

D. $x \neq 2, -1, 4, -3$

9. Numerical Response For all permissible values of x, $\dfrac{5x^2-11x-12}{x^2-x-6} \div \dfrac{x-1}{x^2+x-2}$ simplifies to a linear expression $Ax + B$. The value of $A \times B$ is _____.

C. Stretch

10. Determine the rational expression that belongs in the box: $\square \div \dfrac{x^2-9}{x^2+x} = \dfrac{x}{x-3}$

11. Simplify $\dfrac{x^2-1}{x^2+4x} \div \left(\dfrac{x^2+x}{x^2-16} \div \dfrac{x^2-4x}{x-1} \right)$. State the non-permissible values.

12. Explain why $\dfrac{x-3}{x+1} \div \dfrac{x-3}{x+1}$ equals 1 for most values of x but not for all. For which values is it not equal to 1?

Extra Practice 5: Adding and Subtracting with Monomial Denominators

A. Skills

1. Simplify.

(a) $\frac{7}{12}+\frac{5}{18}$ (b) $\frac{3}{10}-\frac{5}{6}$ (c) $\frac{x-2}{3}+\frac{x+4}{6}$ (d) $\frac{5n+1}{8}-\frac{n-3}{4}$

2. Simplify. State the non-permissible values.

(a) $\frac{9}{4y}-\frac{1}{4y}$ (b) $\frac{2x+9}{x+3}-\frac{x+6}{x+3}$ (c) $\frac{a^2}{a-4}-\frac{16}{a-4}$

3. Simplify. State the non-permissible values.

(a) $\frac{1}{2a}+\frac{3}{5a}$ (b) $\frac{6}{x^2}-\frac{1}{3x}$ (c) $\frac{3}{8m^2n}+\frac{5}{12mn^2}$

(d) $\frac{x-1}{6x}+\frac{x+2}{9x}$ (e) $\frac{4a+1}{2a^2}-\frac{a-3}{3a}$ (f) $3x-2+\frac{1}{x}$

4. Determine the LCD. (a) $9a^3b$ and $6ab^2$ (b) 5, $2x$ and $10x^2$ (c) $4p^2$, $6pq$ and $3q^3$

B. Applying

5. Show that $\frac{a+b}{ab}-\frac{1}{a}=\frac{1}{b}$ for $a\neq 0$, $b\neq 0$.

6. Sam simplified $\frac{4}{3a}-\frac{a+2}{3a}$ as $\frac{6-a}{3a}$. Explain the error and give the correct answer.

7. A triangle has sides $\frac{3}{2x}$, $\frac{5}{3x}$ and $\frac{7}{6x}$ cm, where $x > 0$. Write its perimeter in simplest form.

8. An athlete swims 2 km at s km/h and then runs 10 km at $4s$ km/h.

(a) Write the total time as a single rational expression in simplest form.

(b) Determine the total time when $s = 3$.

9. The expression $\frac{x}{3} - \frac{x-6}{2}$ simplifies to

A. $\frac{-x-18}{6}$ **B.** $\frac{18-x}{6}$ **C.** 6 **D.** $\frac{5x-18}{6}$

10. Numerical Response For $a \neq 0$, $\frac{2a+1}{3a} + \frac{a-2}{6a}$ simplifies to a constant. To the nearest hundredth, the constant is _____.

C. Stretch

11. Determine all pairs of positive integers A and B for which $\frac{A}{2x} + \frac{B}{3x} = \frac{13}{6x}$.

12. Simplify $\dfrac{\frac{x}{y} - \frac{y}{x}}{\frac{1}{x} + \frac{1}{y}}$. State the restrictions.

13. (a) Show that $\frac{1}{2x} + \frac{1}{3x} + \frac{1}{6x} = \frac{1}{x}$, $x \neq 0$.

(b) Use part (a) to write $\frac{1}{5}$ as the sum of three fractions that each have numerator 1.

Extra Practice 6: Adding and Subtracting with Binomial and Trinomial Denominators

A. Skills

1. Simplify. State the non-permissible values.

(a) $\dfrac{6}{x-1}+\dfrac{1}{x+3}$

(b) $\dfrac{3}{2a+1}-\dfrac{2}{a}$

(c) $\dfrac{x-4}{x+2}+\dfrac{x+1}{x-3}$

(d) $\dfrac{2}{x-y}-\dfrac{1}{x+y}$

2. Factor the denominators first. Simplify fully and state the non-permissible values.

(a) $\dfrac{7}{3x-12}-\dfrac{2}{x-4}$

(b) $\dfrac{5}{x+1}-\dfrac{10}{x^2+x}$

3. Simplify fully. State the non-permissible values.

(a) $\dfrac{2}{x^2+2x-3}+\dfrac{1}{x^2-1}$

(b) $\dfrac{5}{x^2+3x-4}+\dfrac{2}{x^2-4x+3}$

4. Simplify $\dfrac{4x}{2x-3}+\dfrac{6}{3-2x}$. State the non-permissible value.

B. Applying

5. Simplify fully: $\dfrac{x^2-9}{x^2+x-6}-\dfrac{1}{x-2}$. State the non-permissible values.

6. A train travels 300 km at an average speed of x km/h, then 200 km through mountains at 20 km/h slower.

(a) Write the total time as a single rational expression in simplest form, and state the values of x that make sense.

(b) Determine the total time when $x = 100$.

7. Leo says $\dfrac{1}{x} + \dfrac{1}{x+1} = \dfrac{2}{2x+1}$. Test his claim with $x = 1$, then simplify the left side correctly.

8. For $x \neq 2, -1$, $\dfrac{x+5}{x^2-x-2} - \dfrac{2}{x-2}$ simplifies to

A. $\dfrac{3-x}{(x-2)(x+1)}$ **B.** $\dfrac{7-x}{(x-2)(x+1)}$ **C.** $\dfrac{x+3}{(x-2)(x+1)}$ **D.** $\dfrac{4-x}{(x-2)(x+1)}$

9. Numerical Response The expression $\dfrac{3}{x-2} + \dfrac{2}{x+3}$ can be written as $\dfrac{Ax+B}{(x-2)(x+3)}$. The value of $A + B$ is _____.

C. Stretch

10. Simplify $\dfrac{\dfrac{x}{x-1} - 1}{\dfrac{1}{x-1} + 1}$. State the non-permissible values.

11. Determine the constants A and B so that $\dfrac{A}{x+1} + \dfrac{B}{x-2} = \dfrac{5x-1}{(x+1)(x-2)}$.

12. (a) Show that $\dfrac{1}{n} - \dfrac{1}{n+1} = \dfrac{1}{n(n+1)}$.

(b) Use part (a) to find the exact sum $\dfrac{1}{1\cdot 2} + \dfrac{1}{2\cdot 3} + \dfrac{1}{3\cdot 4} + \cdots + \dfrac{1}{9\cdot 10}$.

Extra Practice 7: Solving Rational Equations

A. Skills

1. State the NPVs, then solve.

(a) $\frac{8}{x} - 3 = \frac{2}{x}$ (b) $\frac{1}{3x} + \frac{1}{2} = \frac{5}{6x}$ (c) $\frac{x-4}{2x} = \frac{3}{x} - \frac{1}{4}$

2. State the NPVs, then solve.

(a) $\frac{6}{x-2} = \frac{4}{x+1}$ (b) $\frac{x+5}{2x-1} = \frac{x-1}{2x+3}$ (c) $\frac{3}{4-n} = \frac{1}{2}$

3. State the NPVs, solve and verify.

(a) $\frac{5}{x-3} - \frac{2}{x+3} = \frac{24}{x^2-9}$ (b) $\frac{1}{x+4} + \frac{2}{x} = \frac{4}{x^2+4x}$

B. Applying

4. Show that $\frac{2x}{x+4} + 3 = \frac{-8}{x+4}$ has no solution.

5. Solve $\frac{9}{x-5} = 2 + \frac{x-1}{5-x}$.

6. Solve $\frac{x}{x-1} - \frac{3}{x+2} = 1$.

7. Use a graphing calculator to solve $\frac{2}{x+3} = \frac{x}{2} - 1$, to the nearest hundredth.

8. A punch recipe uses juice and soda in the ratio $5 : 3$. A bowl of punch contains 2.4 L more juice than soda. Write and solve a rational equation to find the amount of each.

9. The solution of $\frac{x}{x+2} = \frac{3}{x+2} + 1$ is

A. $x = 3$ **B.** $x = -2$ **C.** $x = 1$ **D.** no solution

10. Numerical Response The solution of $\frac{4}{2x-1} = \frac{3}{x+2}$, to the nearest tenth, is _____.

C. Stretch

11. Consider $\frac{2x+k}{x-1} = 5$.

(a) The solution is $x = 4$. Determine k.

(b) For what value of k does the equation have no solution?

12. The lens equation is $\frac{1}{f} = \frac{1}{p} + \frac{1}{q}$.

(a) Solve for q.

(b) Determine q when $f = 10$ cm and $p = 15$ cm.

13. Solve $\frac{3}{x^2-x-6} - \frac{1}{x^2-4} = \frac{4}{x^2-5x+6}$.

Extra Practice 8: Rational Equations That Lead to Quadratics

A. Skills

1. State the NPVs and solve.

(a) $x + \dfrac{10}{x} = 7$ (b) $\dfrac{3}{x} = \dfrac{x-2}{5}$ (c) $\dfrac{2}{x} + \dfrac{3}{x+1} = 2$

2. State the NPVs and solve. Reject any extraneous root.

(a) $\dfrac{x^2 - x - 6}{x+2} = 2x - 7$ (b) $\dfrac{x}{x-1} + \dfrac{1}{x+2} = \dfrac{3}{x^2+x-2}$

3. Solve $\dfrac{x+9}{x-3} + \dfrac{4}{3-x} = x$.

4. Solve $\dfrac{3}{x} + \dfrac{x}{x-2} = 3$. Give exact roots.

B. Applying

5. Show that $\dfrac{x}{x+4} + \dfrac{4}{x} = \dfrac{16}{x^2+4x}$ has no solution.

6. Without fully solving, explain why $\dfrac{x^2}{x-3} = \dfrac{9}{x-3}$ has exactly one solution, and state it.

7. Two resistors are connected in parallel ($\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$). One resistor is 6 Ω greater than the other, and the combined resistance is 4 Ω. Determine both resistances.

8. The solutions of $\frac{6}{x} - \frac{1}{x+1} = 1$ are

A. $2 \pm \sqrt{10}$ **B.** $-2 \pm \sqrt{10}$ **C.** $4 \pm 2\sqrt{10}$ **D.** $2 \pm \sqrt{22}$

9. Numerical Response The positive solution of $\frac{1}{x} + \frac{1}{x+2} = \frac{1}{2}$, to the nearest hundredth, is _____.

C. Stretch

10. The equation $\frac{3}{x} + \frac{x}{x-2} = k$ has the root $x = 3$. Determine k and the other root.

11. Determine all values of k for which $x + \frac{k}{x} = 4$ has no real solution.

12. Solve $\frac{x+1}{x-1} - \frac{x-1}{x+1} = \frac{8}{3}$.

Extra Practice 9: Problems Involving Rational Equations (work, motion, numbers)

A. Skills

1. Write an expression for each quantity.

(a) the time to drive 150 km at $(s + 10)$ km/h

(b) the upstream speed of a boat with still-water speed s km/h in a 3 km/h current

(c) the fraction of a job done in 4 h by a worker who needs x h alone

(d) the sum of a number and its reciprocal

2. A number minus 15 times its reciprocal is 2. Determine the number.

3. A walker covers 9 km in the same time that a cyclist covers 24 km. The cyclist is 10 km/h faster. Determine both speeds.

4. Ana can mow a lawn in 30 min and Ben can mow it in 60 min. How long does it take them together?

B. Applying

5. A car makes a 180 km trip 1 hour faster than a bus, because its average speed is 15 km/h greater. Determine the speed of each.

6. A jet boat travels at 30 km/h in still water. It goes 36 km up a river and 36 km back in 2.5 h. Determine the speed of the current.

7. A small pipe takes 6 h longer than a large pipe to fill a tank. Together they fill it in 4 h. How long does each pipe take alone?

8. The sum of the reciprocals of two consecutive even integers is $\frac{5}{12}$. Determine the integers.

9. A plane with airspeed s km/h flies 600 km into a 25 km/h headwind. The time for the flight, in hours, is

A. $\frac{600}{s+25}$ **B.** $\frac{s-25}{600}$ **C.** $600(s-25)$ **D.** $\frac{600}{s-25}$

10. Numerical Response Jess cycles 45 km to a lake and returns at a speed 3 km/h slower. The round trip takes 5.5 h of cycling. Her speed going to the lake, in km/h, is _____.

C. Stretch

11. A plane flies 1200 km into a 40 km/h headwind and then returns with the same wind as a tailwind. The two flights take 12.5 h in total. Determine the plane's airspeed.

12. Crew A can pave a parking lot in 12 days. Crew A works alone for 3 days, then Crew B joins, and together they finish in 5 more days. How long would Crew B take alone?

13. A 40 L tank of coolant is 25% antifreeze. How many litres of pure antifreeze must be added to make the mixture 40% antifreeze?

14. Owen cycles 30 km to a friend's house and returns 5 km/h faster. Each way is the same route, and the round trip takes 3 h of riding. Determine his speed each way, to the nearest tenth of a km/h.

Extra Practice 10: Unit 6 Review — Practice Test

State the non-permissible values for every expression and equation. Give answers in simplest form.

A. Multiple Choice

1. The non-permissible values of $\dfrac{x-2}{x^2+2x-15}$ are

A. $2, -5, 3$ **B.** $-5, 3$ **C.** $5, -3$ **D.** 2 only

2. In simplest form, $\dfrac{x^2-7x+10}{x^2-4}$ is

A. $\dfrac{x-5}{x+2}$ **B.** $\dfrac{x+5}{x-2}$ **C.** $\dfrac{-7x+10}{-4}$ **D.** $\dfrac{x-5}{x-2}$

3. In simplest form, $\dfrac{8-2x}{x^2-16}$ is

A. $\dfrac{2}{x+4}$ **B.** $-\dfrac{2}{x-4}$ **C.** $-\dfrac{2}{x+4}$ **D.** $\dfrac{2}{x-4}$

4. When $\dfrac{x^2-9}{x^2-3x}$ is simplified to $\dfrac{x+3}{x}$, the non-permissible values are

A. $x \neq 0$ only **B.** $x \neq 0, 3$ **C.** $x \neq -3, 0$ **D.** $x \neq \pm 3, 0$

5. The product $\dfrac{x^2+x-6}{x^2-4x} \cdot \dfrac{x^2-16}{x^2+7x+12}$ simplifies to

A. $\dfrac{x+2}{x}$ **B.** $\dfrac{x}{x-2}$ **C.** $\dfrac{(x-2)(x+4)}{x}$ **D.** $\dfrac{x-2}{x}$

6. The non-permissible values of $\dfrac{x}{x+2} \div \dfrac{x-5}{x-1}$ are

A. $-2, 1$ only **B.** $-2, 5$ only **C.** $-2, 1, 5$ **D.** $0, -2, 1, 5$

7. The sum $\dfrac{2}{3x} + \dfrac{1}{x^2}$ is

A. $\dfrac{3}{3x+x^2}$ **B.** $\dfrac{2x+3}{3x^2}$ **C.** $\dfrac{3}{3x^3}$ **D.** $\dfrac{5}{3x^2}$

8. The difference $\dfrac{3}{x+2} - \dfrac{x-1}{x+2}$ is

A. $\dfrac{4-x}{x+2}$ **B.** $\dfrac{2-x}{x+2}$ **C.** $\dfrac{x-4}{x+2}$ **D.** $\dfrac{4-x}{0}$

9. The solution of $\dfrac{x}{x-3} = \dfrac{3}{x-3} + 2$ is

A. $x = 3$ **B.** $x = -3$ **C.** $x = 0$ **D.** no solution

10. Rae can paint a room in 5 h. With Tom helping, it takes 3 h. If Tom alone takes t hours, an equation for t is

A. $\frac{3}{5}+\frac{3}{t}=1$ **B.** $\frac{1}{5}+\frac{1}{t}=3$ **C.** $5+t=3$ **D.** $\frac{5}{3}+\frac{t}{3}=1$

B. Numerical Response

11. Numerical Response The difference $\frac{4}{x-3}-\frac{2}{x+1}$ can be written as $\frac{ax+b}{(x-3)(x+1)}$. Write the value of a in the first box and the value of b in the next two boxes.

12. Numerical Response A positive number is 3 more than 10 times its reciprocal. The number is _____.

13. Numerical Response With a 50 km/h tailwind a plane flies 900 km in the same time it flies 700 km into the same wind. The plane's airspeed, in km/h, is _____.

14. Numerical Response The positive root of $\frac{2}{x}+\frac{3}{x+1}=1$, to the nearest hundredth, is _____.

C. Written Response

15. Simplify and state the non-permissible values.

(a) $\frac{x^2-2x-3}{x^2-9}\div\frac{x^2-1}{x^2+2x-3}$

(b) $\frac{x+4}{x^2+x-2}-\frac{2}{x^2-1}$

16. Solve $\dfrac{x}{x+2} - \dfrac{1}{x-2} = \dfrac{8}{x^2-4}$ algebraically. State the non-permissible values, identify any extraneous root and verify the solution.

17. A tour bus travels 280 km from one city to another. On the return trip it averages 10 km/h more and takes 30 minutes less.

(a) If the speed on the first trip is s km/h, write expressions for the time of each trip.

(b) Determine the speed of each trip.

(c) Determine the total driving time for the round trip.

18. Working alone, Crew B takes 10 days longer than Crew A to frame a house. Working together, the crews take 12 days. Determine how long each crew takes alone, and explain why one root of your equation is rejected.